

Histogram Matching Numerical

Q. Perform histogram mapping in given image data with reference histogram.

Image data

r	0	1	2	3	4	5	6	7
n	45	62	12	53	17	23	34	66

Reference histogram

r	0	1	2	3	4	5	6	7
n	0	0	0	45	23	65	92	87

Step-1: Calculate the CDF of given image data. (Ha)

r	n	PDF	CDF
0	45	0.14423077	0.14423077
1	62	0.19871795	0.34294872
2	12	0.03846154	0.38141026
3	53	0.16987179	0.55128205
4	17	0.05448718	0.60576923
5	23	0.07371795	0.67948718
6	34	0.10897436	0.78846154
7	66	0.21153846	1

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Step-2: Calculate the CDF of reference histogram. (Hb)

r	n	PDF	CDF
0	0	0	0
1	0	0	0
2	0	0	0
3	45	0.14423077	0.14423077
4	23	0.07371795	0.21794872
5	65	0.20833333	0.42628205
6	92	0.29487179	0.72115385
7	87	0.27884615	1

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Step-3: Matching table

r(Ha)	CDF(Ha)	CDF(Hb)	s
0	0.14423077	0	3
1	0.34294872	0	5
2	0.38141026	0	5
3	0.55128205	0.14423077	5
4	0.60576923	0.21794872	6
5	0.67948718	0.42628205	6
6	0.78846154	0.72115385	6
7	1	1	7

For matched image histogram.

r(Ha)	n(Ha)	s	n(s)
0	45	3	45
1	62	5	127
2	12	5	
3	53	5	
4	17	6	74
5	23	6	
6	34	6	
7	66	7	66
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Step-4: Display image histogram after matching

r	0	1	2	3	4	5	6	7
n	0	0	0	45	0	127	74	66